PROJECT REPORT OF CO-OP Project at Industry (Module-2)

ON

Detecting Hallucinations in LLMs using Prompt Validation and Response Scoring

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BACHELOR OF ENGINEERING

In

COMPUTER SCIENCE AND ENGINEERING

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DECLARATION

I hereby certify that the work which is being presented in the project report entitled "Detecting Hallucinations in LLMs using Prompt Validation and Response Scoring" in partial fulfilment of requirement for the award of the degree of Bachelor of Engineering (Computer Science and Engineering) submitted in the department of Computer Science and Engineering at Chitkara University Institute of Engineering and Technology, Chitkara University, Punjab, India, is an authentic record of our own work carried out under the supervision of Dr. Swati Goel. The matter presented in this project report has not been submitted in any other university/institute for the award of any degree.

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This is to certify that the above statement made by the candidate is correct to the best of my knowledge and belief.

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I would also like to thank my teammates and peers for their cooperation, discussions, and helpful suggestions which contributed positively to the completion of this research work.

ABSTRACT

Large Language Models (LLMs) have become an important part of modern artificial intelligence due to their ability to generate meaningful and human-like text. These models are widely used in applications such as content generation, chatbots, text summarization, question answering, and research assistance. However, despite their capabilities, LLMs often generate incorrect or misleading information, commonly known as hallucinations. Such responses may appear correct but are not supported by reliable facts, which affects the trustworthiness of AI systems.

This project focuses on reducing hallucinations in LLMs through prompt validation and response scoring techniques**.** The study analyzes how different prompt engineering methods can improve the reliability and factual accuracy of model-generated responses. A structured evaluation was conducted using different categories of prompts, including factual, logical, and descriptive prompts. Initially, responses generated through basic prompts were analyzed and compared with responses generated using refined prompts containing explicit instructions, contextual guidance, and verification steps.

The findings show that prompt engineering techniques significantly reduce hallucinated outputs and improve response quality. The study highlights that carefully designed prompts can guide language models toward generating more reliable and contextually accurate responses. The proposed approach provides a simple and cost-effective method to improve the performance of AI systems without requiring major changes to the underlying model.

1. Introduction

1.1 Background of the Study

In recent years, Large Language Models (LLMs) have transformed the field of artificial intelligence and natural language processing. These models are capable of understanding and generating human-like text by learning from large amounts of training data. Applications such as virtual assistants, content generation systems, chatbots, educational platforms, and automated customer support widely depend on LLMs for generating meaningful responses.

Although these models are highly advanced, they often face a major issue known as hallucination. Hallucination occurs when a language model generates information that appears correct and convincing but is actually false, inaccurate, or unsupported by facts. This creates a challenge in fields where factual correctness is important, such as healthcare, education, legal systems, and academic research.

As the use of AI systems continues to grow, improving the reliability and trustworthiness of generated responses has become necessary. One of the effective approaches to reduce hallucinations is prompt engineering, where prompts are carefully designed to guide the model toward more accurate and contextually relevant outputs.

This project focuses on analyzing different prompt engineering strategies to reduce hallucinated responses generated by Large Language Models. The research evaluates whether structured prompts, contextual information, and response validation techniques can improve the factual consistency of AI-generated outputs.

1.2 Problem Statement

Large Language Models are capable of generating fluent and meaningful text, but they sometimes provide false or misleading information with high confidence. These hallucinated responses reduce user trust and affect the reliability of AI systems, especially in domains where accuracy is essential. Therefore, there is a need to explore techniques that can reduce hallucinations and improve the correctness of generated responses.

1.3 Aim of the Project

The main aim of this project is to study and reduce hallucinations in Large Language Models using prompt validation and response scoring techniques.

1.4 Objectives of the Project

The objectives of this project are:

- To understand the problem of hallucinations in Large Language Models.
- To analyze the impact of prompt engineering on response quality.
- To compare responses generated using normal prompts and refined prompts.
- To reduce hallucinated outputs by using structured and validated prompts.
- To improve the reliability and factual accuracy of AI-generated content.

1.5 Scope of the Project

This project focuses on studying hallucinations generated by Large Language Models and reducing them using prompt engineering techniques. The research mainly emphasizes factual correctness and response reliability. The methods used in this study can be useful in improving AI-based applications used in education, research, content generation, and customer support systems.

2. Methodology

2.1 Overview

The methodology of this project focuses on understanding and reducing hallucinations in Large Language Models (LLMs) through prompt engineering techniques. The research was conducted by analyzing how different prompt structures affect the quality and factual accuracy of generated responses.

The study followed a systematic process that included baseline testing, prompt refinement, response collection, and evaluation. Different prompts were designed and tested to observe whether modifications in prompt structure could reduce hallucinated responses and improve the reliability of outputs generated by language models.

The complete process involved generating responses using standard prompts and then comparing them with responses generated using improved prompts that contained better instructions, context, and validation methods.

2.2 Research Approach

The research was conducted using an experimental approach to examine the effectiveness of prompt engineering in reducing hallucinations. Different prompts were created and categorized based on factual knowledge, logical reasoning, and descriptive content generation.

Initially, normal prompts were given to the language model to establish a baseline understanding of hallucination frequency. Later, prompt refinement techniques were introduced to observe whether the generated responses became more accurate and reliable.

The overall study focused on comparing outputs before and after applying prompt modifications.

2.3 Experimental Process

The methodology was divided into the following stages:

2.3.1 Baseline Assessment

In the first stage, general prompts were provided to the language model without adding any specific instructions or constraints. This helped in identifying the initial hallucination rate and understanding how the model responded under normal conditions.

The responses collected during this phase were evaluated to identify incorrect, misleading, or unsupported information.

2.3.2 Prompt Refinement Phase

After baseline assessment, prompts were modified using different prompt engineering strategies to reduce hallucinations. The following techniques were applied:

a) Contextual Prompting

In this technique, additional background information was added to prompts to guide the model more effectively. Providing context helped the model understand the topic more clearly and generate relevant responses.

b) Structured Prompting

The prompts were organized in a structured format using questions, lists, and direct instructions. This reduced ambiguity and guided the model toward focused answers.

c) Explicit Instruction-Based Prompting

Specific instructions such as “Provide factual information only” or “Avoid assumptions” were included in prompts to improve response accuracy and reduce misinformation.

d) Verification-Oriented Prompting

Prompts were designed in a way that encouraged the model to verify uncertain information and avoid unsupported claims.

2.4 Data Collection

The data for this project was collected by submitting different prompts to the language model and recording the generated responses. Multiple categories of prompts were used to ensure diverse testing conditions.

The collected responses were carefully examined to identify hallucinations and compare the effectiveness of different prompt strategies.

The prompts used in this research included:

- Fact-based prompts
- Logical reasoning prompts
- Descriptive content prompts
- Contextual prompts
- Verification-based prompts

2.5 Evaluation Process

The generated responses were evaluated based on their factual correctness, consistency, and relevance.

The main evaluation criteria included:

- Accuracy: Whether the response contained factual information.
- Consistency: Whether the generated content remained relevant to the query.
- Reliability: Whether the response could be trusted without misinformation.
- Hallucination Rate: The percentage of responses containing false or unsupported information.

A comparison was performed between normal prompts and refined prompts to observe the reduction in hallucinations.

2.6 Validation of Results

To improve the reliability of findings, multiple evaluations were conducted on generated responses. Responses were checked for consistency and compared using different prompt formats.

The study ensured that prompt engineering techniques were tested repeatedly to confirm whether they genuinely improved response quality and reduced hallucinated outputs.

3. Tools And Technologies

The successful completion of this project required the use of various tools and technologies for conducting research, prompt testing, response analysis, and documentation.

3.1 Large Language Models (LLMs)

Large Language Models were the primary technology used in this research. These models were tested to analyze how prompt modifications influence generated outputs and reduce hallucinations.

The responses generated by the language models were evaluated for factual correctness and reliability.

3.2 Prompt Engineering Techniques

Prompt engineering played an important role in this project. Different prompt design methods were applied to improve response quality and reduce hallucinated outputs.

Some techniques used include:

- Contextual Prompting
- Structured Prompting
- Explicit Instruction-Based Prompting
- Fact-Based Prompting
- Verification-Based Prompting

3.3 Python

Python was used for handling research-related analysis and organizing collected responses. Python is widely preferred because of its simplicity and availability of libraries for artificial intelligence and data processing.

3.4 Microsoft Word

Microsoft Word was used for preparing and formatting the research report and documentation according to the university guidelines.

3.5 Research Papers and Academic Sources

Various research papers, journals, and scholarly articles were studied to understand hallucinations in LLMs and existing techniques used for mitigation. These references helped in building the foundation of this research work.

4. Implementation

4.1 Project Implementation Overview

The implementation of this project involved testing and analyzing different prompt engineering methods to reduce hallucinations in Large Language Models.

The process began with creating a set of prompts that covered different categories such as factual questions, logical reasoning, and descriptive tasks. These prompts were first tested using standard prompting methods to understand the initial behavior of the language model.

After this, prompts were modified using different engineering techniques to improve response quality.

4.2 Prompt Design Process

Different prompt structures were carefully designed to test their effectiveness in reducing hallucinations.

The implementation included:

Step 1: Creating Baseline Prompts

Simple prompts were created and tested without adding instructions or context. The responses generated during this phase were recorded and analyzed.

Step 2: Refining Prompts

The prompts were improved by adding context, instructions, and verification-related conditions.

For example:

- Prompts were made more specific.
- Background information was added.
- Instructions for factual correctness were included.
- Ambiguous wording was avoided.

Step 3: Response Collection

The generated outputs from different prompt versions were collected and stored for comparison.

Step 4: Response Evaluation

The collected responses were examined to determine whether they contained hallucinated or unsupported information.

The performance of refined prompts was compared with baseline prompts to identify improvements.

4.3 Response Scoring

A response scoring mechanism was used to evaluate generated outputs based on accuracy and reliability.

Responses were analyzed based on:

- Correctness of facts
- Relevance to the query
- Clarity of response
- Presence of hallucinated information

This scoring process helped in understanding which prompting strategy performed better.

4.4 Performance Analysis

After implementation, it was observed that refined prompts generated more accurate and reliable outputs compared to normal prompts. Prompt engineering significantly reduced misinformation and improved response consistency.

5. Results

5.1 Overview of Results

The main objective of this project was to analyze whether prompt engineering techniques can reduce hallucinations in Large Language Models (LLMs) and improve the reliability of generated responses. After conducting multiple experiments using different prompt structures, noticeable improvements were observed in response quality and factual accuracy.

The study showed that the way prompts are written has a direct impact on the responses generated by language models. Refined prompts with proper instructions and contextual guidance produced more reliable and accurate outputs than basic prompts.

5.2 Reduction in Hallucination Rate

During the initial testing phase, the language model generated responses using normal prompts without any additional guidance. These responses contained several factual inaccuracies and unsupported statements.

After applying prompt engineering techniques such as contextual framing, structured prompting, and explicit instructions, the number of hallucinated responses decreased significantly.

The findings indicated that:

- Refined prompts generated more factually correct responses.
- Clear and structured instructions reduced ambiguity in outputs.
- Context-based prompts helped the model stay relevant to the topic.
- Verification-focused prompts reduced unsupported claims.

The hallucination rate showed a noticeable reduction after prompt refinement, proving that prompt design plays an important role in improving response quality.

5.3 Comparative Analysis of Prompting Techniques

Different prompt engineering methods were tested and compared during the study.

a) Closed-Ended Prompting

This technique generated the best results because it restricted the model to more direct and specific responses. Since the model had limited room for assumptions, hallucinations were reduced considerably.

b) Contextual Prompting

Providing additional context improved response relevance and accuracy. The model generated better responses when sufficient background information was included.

c) Fact-Based Prompting

Including factual details such as names, dates, or specific information helped guide the model toward more reliable outputs.

d) Explicit Instruction Prompting

Prompts that clearly instructed the model to avoid assumptions and provide factual responses performed better than general prompts.

5.4 Key Outcomes of the Project

The following outcomes were achieved through this research:

- Better understanding of hallucinations in LLMs.
- Identification of prompt engineering as an effective mitigation technique.
- Improved response reliability through refined prompts.
- Reduction in misleading or unsupported information generated by AI systems.
- Development of a structured approach for prompt validation and response scoring.

5.5 Practical Significance of the Study

The findings of this project can be useful in real-world applications where response accuracy is important. These include:

- Educational systems

- Research platforms
- AI-based content generation tools
- Customer support systems
- Information retrieval applications

6. Conclusion And Future Scope

6.1 Conclusion

This project focused on reducing hallucinations in Large Language Models (LLMs) using prompt validation and response scoring techniques. Hallucinations are one of the major challenges in language models because they generate information that may sound correct but is actually inaccurate or unsupported.

Through this study, it was observed that prompt engineering plays an important role in improving the reliability and factual correctness of model-generated responses. Different prompt techniques such as contextual prompting, structured prompting, fact-based prompting, and explicit instruction-based prompting were tested and analyzed.

The results showed that refined prompts significantly reduced hallucinations and improved the consistency of responses. Properly designed prompts helped guide the language model toward generating more relevant and trustworthy information.

Overall, the study concludes that prompt engineering is a simple, effective, and cost-efficient method for improving the quality of responses generated by LLMs. It can help increase trust in AI systems, especially in fields where accuracy is important.

6.2 Future Scope

Although this project successfully reduced hallucinations to some extent, there are still several opportunities for future improvement.

Future work can focus on:

- Developing automated prompt generation techniques.
- Combining prompt engineering with external fact-checking systems.
- Testing hallucination reduction techniques on more advanced language models.
- Applying the approach in domain-specific areas such as healthcare, law, and education.
- Integrating real-time verification systems to improve response accuracy.

In the future, as AI systems continue to evolve, more advanced techniques can be developed to make language models more reliable, transparent, and trustworthy.

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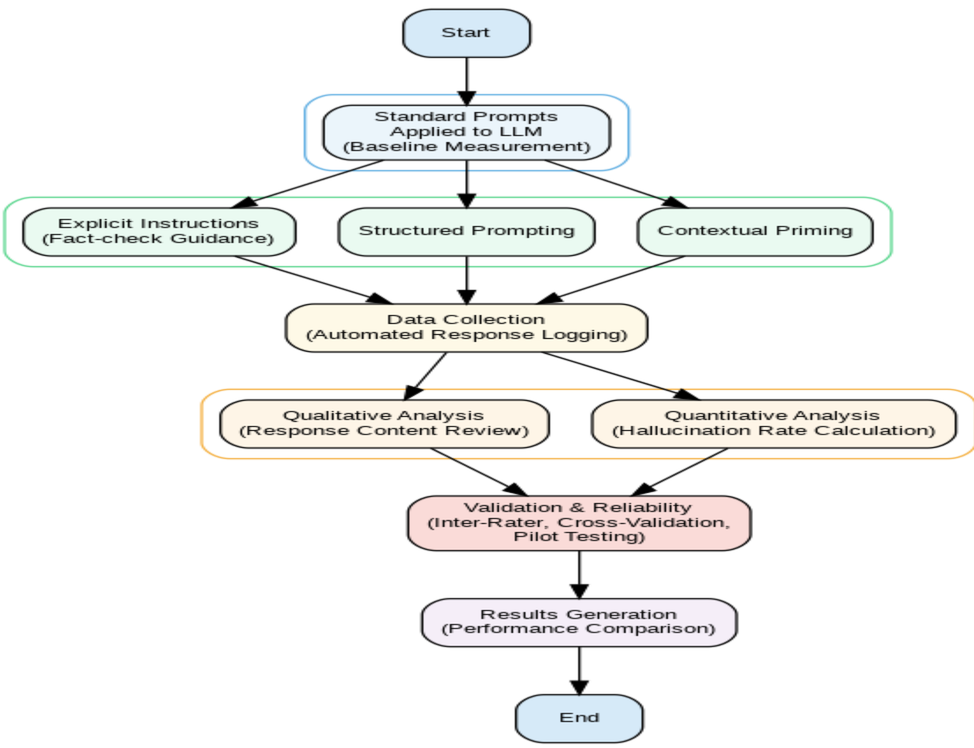
8. Appendices

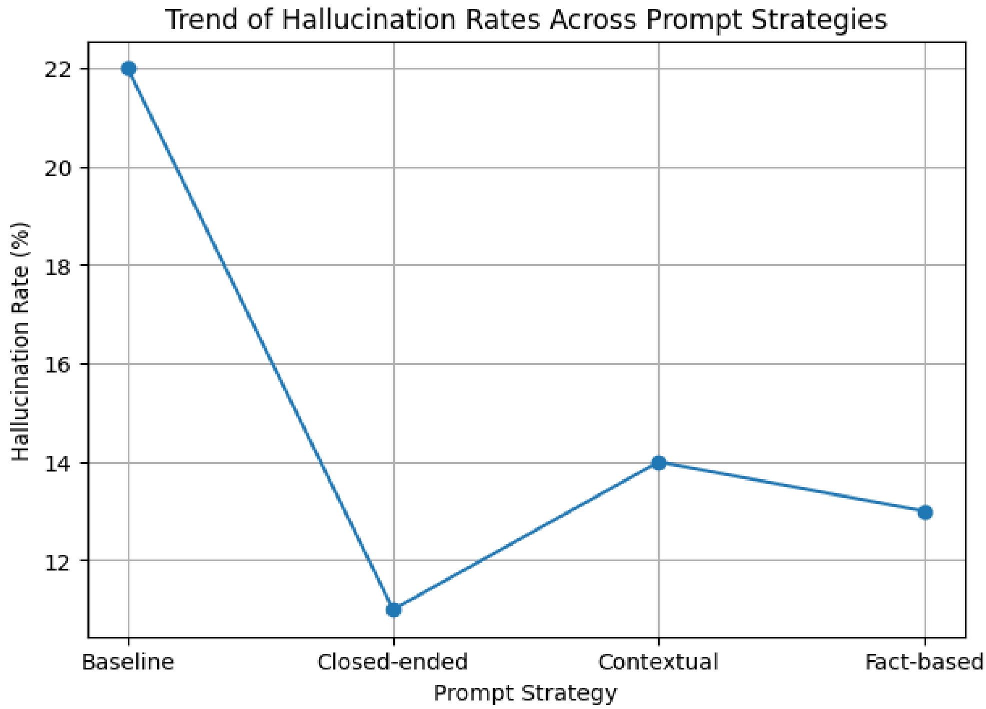
Appendix A: Methodology Diagram

This below Flowchart shows the methodology used for prompt refinement, response collection, and evaluation.



Appendix B: Result Graph

This below Graph shows the comparison of results and hallucination reduction after applying prompt engineering techniques.


Trend of Hallucination Rates Across Prompt Strategies
Hallucination Rate (%)
22
20
18
16
14
12
Baseline
Closed-ended
Contextual
Fact-based
Prompt Strategy